

Reduction #3: Max-of- K Correlated Gaussian Fields and the Extreme-Value Origin of Tempotron Fragmentation

Tempotron learning-surface program

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Abstract

The tempotron’s binary decision is the sign of its maximum membrane potential over time, which reduces in the large- N limit to the maximum of K correlated standard Gaussian fields. Extreme-value theory (EVT) then yields a *leading-order, annealed* formula (prefactor undetermined) for the expected number of disconnected solution clusters as a function of load α and field count K :

$$N_{\text{clusters}}(\alpha, K) \approx \exp[N S_{\text{cl}}(\alpha, K)] = (\ln K)^{N/2} \cdot 2^{-N\alpha},$$

where the equality holds to leading order in $\ln K$ (an undetermined $O(1)$ prefactor is absorbed in the subleading $\frac{N}{2} \ln c$ term; see §3.3). Setting $N_{\text{clusters}} = 1$ recovers the capacity lift $\alpha_c(K) = (\ln \ln K)/(2 \ln 2)$, and the inter-cluster overlap $q_0 \sim \alpha / \ln K \rightarrow 0$ quantifies the fragmentation of the solution manifold. The core result—lift and shattering are one EVT effect counted two ways—is verified at three independent points: the Gumbel scale, the $K = 1$ perceptron limit, and the $S_{\text{cl}} = 0$ capacity crossing. An independent codex cross-check [9] shows that the commonly-cited additive offset $\alpha_0 \approx 2.58$ is a small- K fit artifact: the first-moment calculation gives a third iterated-log subleading term, not a constant. Status: **PARTIAL**—leading scalings solid; $O(1)$ constants and quenched cluster complexity open.

Contents

1	Introduction	2
2	Setup and Assumptions	3
3	Derivation	3
3.1	EVT statistics of the maximum of K Gaussians	3
3.2	Intra-cluster scale from pairwise agreement	4
3.3	Volume bookkeeping and the cluster count	4
3.4	The expected number of clusters	5
3.5	Capacity from the cluster count	5
3.6	Leading-order scaling of inter-cluster overlap q_0	5
4	Results Summary	6
5	Independent Verification	6
6	Experimental Test: the Solution Space Shatters	7
7	Limitations and Open Problems	8

8	Discussion	9
9	Methods	10
9.1	What is measured: q_0 as a Monte-Carlo estimator of inter-solution overlap	10
9.2	Model and runs reused from reduction #1	11
9.3	Reference scales: the orthogonality floor and the single-cluster anchor	11
9.4	Controls	12
9.5	Link to reduction #1: capacity and geometry on the same runs	12
9.6	Compute and cost	12
9.7	Finite- N ladder: q_0 is intensive (rises with N)	13
9.8	Caveats	14

Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Introduction

The tempotron [8] is a leaky integrate-and-fire neuron trained to fire (or remain silent) in response to an N -afferent spike-timing pattern. Its membrane potential at time t is

$$V(t; \mathbf{w}) = \sum_{i=1}^N w_i K(t - s_i), \quad (1)$$

where w_i are synaptic weights and K is a double-exponential kernel. The neuron fires if $\max_t V(t; \mathbf{w})$ exceeds a fixed threshold ϑ . Because the kernel is shared across synapses, the time-dependent maximum depends on the same weight vector for all t ; this shared-weight structure is what makes the problem interesting and hard.

Rubin, Monasson, and Sompolinsky [1] showed that in the large- N limit each time-bin field $U_\ell(\mathbf{w})$ is approximately Gaussian (CLT over afferents), so the tempotron’s decision reduces to the sign of

$$S(\mathbf{w}) = \max_{\ell=1, \dots, K} U_\ell(\mathbf{w}), \quad U_\ell \sim \mathcal{N}(0, 1) \text{ correlated}. \quad (2)$$

This reduction—the *max-of- K -correlated-Gaussian-fields* model—is the subject of the present paper.

The central insight is classical: the maximum of K weakly-correlated standard Gaussians converges to a Gumbel distribution [4, 5] with location $\sqrt{2 \ln K}$ and scale $\beta_K = 1/\sqrt{2 \ln K}$. As K grows, the top fields crowd into a window of width $\sim \beta_K \rightarrow 0$. This crowding simultaneously

1. shrinks the radius of each solution cluster to $1 - q_1 = O(1/\ln K)$, generating *exponentially many* clusters; and
2. lifts the capacity above the perceptron baseline by $(\ln \ln K)/(2 \ln 2)$.

These are two descriptions of the same EVT effect.

Contribution. We make the Rubin et al. argument quantitative: we derive N_{clusters} as an explicit function of α and K (Section 4), give the leading-order scaling $q_0 \sim \alpha/\ln K$ of the inter-cluster overlap, prove the $K = 1$ perceptron limit self-consistently (Section 5), and flag the codex finding that no finite additive constant α_0 exists at leading order (Section 7). The derivation rests on the annealed (first-moment) cluster count; the quenched complexity and $O(1)$ constants remain open.

2 Setup and Assumptions

Model. Store $p = \alpha N$ binary patterns $\{\xi_i^\mu\}$ with labels $y^\mu \in \{+1, -1\}$. The weight vector $\mathbf{w}$ lies on the N -sphere. The tempotron score is Equation (2), where each field is

$$U_\ell(\mathbf{w}) = \frac{1}{\sqrt{N}} \sum_{i=1}^N w_i \xi_{i\ell}^\mu,$$

and K is the effective number of independent time bins ($K_{\text{eff}} \approx T/\tau$, where T is the integration window and τ is the kernel decay time).

Assumptions.

- (A1) **Gaussianity.** Each U_ℓ is Gaussian by the CLT over $N \gg 1$ afferents. Requires not-too-heavy inter-bin correlations. This is the step where finite- N starvation of the $\ln \ln K$ lift is concentrated (see Section 7).
- (A2) **Weak correlation / effective independence.** The autocorrelation $C(t) = \mathbb{E}[U(t')U(t' + t)]$ decays to 0 such that Berman's condition $C(t) \ln t \rightarrow 0$ holds [5]. Under (A2) the K bins behave for EVT purposes like K asymptotically independent extremes. Strong cross-bin correlation reduces K below T/τ .
- (A3) **Threshold at median.** The decision threshold $\vartheta = U_{\text{th}}$ is the median of $\max_\ell U_\ell$, so a random pattern fires with probability $\frac{1}{2}$ (the symmetric “fair-coin” convention). This forces the per-pattern volume factor $\frac{1}{2}$ in Section 3.
- (A4) **Two-overlap replica structure.** The cluster geometry is captured by exactly two order parameters: inter-cluster overlap q_0 and intra-cluster overlap q_1 . Justified by the large- K binary-overlap structure ($q \approx 0$ or $q \approx 1$); RSB touches only subleading corrections to α_c [1].
- (A5) **Annealed cluster count.** “ N_{clusters} ” denotes the *expected* (annealed) number of clusters, $\mathbb{E}[N_{\text{cl}}] = V/V_{\text{cl}}$, not the quenched $\mathbb{E}[\ln N_{\text{cl}}]/N$. For the binary-overlap regime these coincide to leading order in N ; the distinction is flagged in Section 7.

3 Derivation

3.1 EVT statistics of the maximum of K Gaussians

For K weakly-correlated standard Gaussian fields satisfying (A1)–(A2), the maximum converges in distribution to a Gumbel law [4]:

$$\max_\ell U_\ell \xrightarrow{d} U_{\text{th}} + \beta_K (X + \ln \ln 2), \quad X \sim \text{Gumbel}, \quad (3)$$

where the Gumbel distribution function is $G(x) = \exp(-e^{-x})$, and

$$U_{\text{th}} = \sqrt{2 \ln K} + O\left(\frac{1}{\sqrt{\ln K}}\right), \quad (4)$$

$$\beta_K = \frac{1}{\sqrt{2 \ln K}} + O\left(\frac{1}{\ln K}\right). \quad (5)$$

The Gumbel median is $-\ln \ln 2 \approx 0.3665$, so the median of $\max_\ell U_\ell$ equals U_{th} , consistent with assumption (A3).

The key observation is that $\beta_K \rightarrow 0$ as $K \rightarrow \infty$: the top fields crowd into a window of width $O(1/\sqrt{\ln K})$ around the threshold.

3.2 Intra-cluster scale from pairwise agreement

Consider two weight vectors with overlap $q = \mathbf{w}^1 \cdot \mathbf{w}^2 / N$. For a positive pattern ($y^\mu = +1$), the tempotron's potential peaks at some bin ℓ^* with $U_{\ell^*} \approx U_{\text{th}}$ (by Equation (3)). Conditioned on $U_{\ell^*}^1$, the second vector's field at ℓ^* is Gaussian with

$$\mathbb{E}[U_{\ell^*}^2 | U_{\ell^*}^1] = q U_{\ell^*}^1, \quad \text{Std}[U_{\ell^*}^2 | U_{\ell^*}^1] = \sqrt{1 - q^2}.$$

The two vectors agree on this pattern only if the gap $(1 - q)U_{\text{th}} \approx (1 - q)\sqrt{2 \ln K}$ is not large compared to the $O(1)$ fluctuation. Joint agreement on all αN patterns tightens a single-pattern tolerance from $1 - q \lesssim 1/\sqrt{\ln K}$ to

$$1 - q_1 = \frac{c}{\ln K}, \quad c = O(1), \quad (6)$$

matching Rubin et al. [1], eq. (5). The explicit prefactor c is not derived here (it requires a large-deviation argument over all αN patterns; see open problem O3 in §7); the replica saddle-point (Section 3.6, Equation (14)) gives $c = 2\alpha$ at leading order, but the derivation of that identification is not reproduced.

Reading. Weight vectors closer than $1 - q_1 = O(1/\ln K)$ classify identically and form one solution cluster. For $q \ll 1 - q_1$ there is no entropic incentive to share overlap; entropy pressure therefore drives the typical cross-cluster overlap to $q_0 \approx 0$.

3.3 Volume bookkeeping and the cluster count

Two volumes on the N -sphere, under assumptions (A3) and (A5):

Total solution volume. Because $q_0 \approx 0$, the αN constraints (fire / be silent) are approximately independent Gaussian events of probability $\frac{1}{2}$ each:

$$\ln V = N\alpha \ln \frac{1}{2} = -N\alpha \ln 2. \quad (7)$$

One cluster's volume. A cluster is the set of weight vectors within intra-cluster overlap q_1 of a solution. On the N -sphere, the cap subtended by $1 - q_1 \approx c/\ln K$ (with c a *dimensionless* $O(1)$ constant from the replica saddle; K itself is a dimensionless count, so $c/\ln K$ is dimensionless and the argument of $\ln$ is well-defined) satisfies

$$\ln V_{\text{cl}} = \frac{N}{2} \ln(1 - q_1) = \frac{N}{2} \ln \frac{c}{\ln K} = -\frac{N}{2} \ln \ln K + \frac{N}{2} \ln c. \quad (8)$$

The term $\frac{N}{2} \ln c$ is an undetermined $O(N)$ contribution that shifts the cluster entropy density by an additive $O(1)$ constant; it is the precise source of the open $O(1)$ prefactor in N_{clusters} (see §7, O3).

Cluster entropy density. The rate function $S_{\text{cl}} = (\ln V - \ln V_{\text{cl}})/N$ is

$$\boxed{S_{\text{cl}}(\alpha, K) = \frac{1}{2} \ln \ln K - \alpha \ln 2} + \text{subleading terms in } K. \quad (9)$$

This is Rubin et al. [1], eq. (6), recovered from first principles.

3.4 The expected number of clusters

Result 1 (Cluster count). Under assumptions (A1)–(A5) the expected number of solution clusters is

$$N_{\text{clusters}}(\alpha, K) = \exp[N S_{\text{cl}}(\alpha, K)] = (\ln K)^{N/2} \cdot 2^{-N\alpha}. \quad (10)$$

The leading scalings are transparent: at fixed α , N_{clusters} grows *super-polynomially* in K as $(\ln K)^{N/2}$ —the analytic origin of “exponentially many small clusters.” At fixed K it decays as $2^{-N\alpha}$: each stored pattern halves the per-weight solution volume.

3.5 Capacity from the cluster count

Solutions exist while at least one cluster survives, $N_{\text{clusters}} \geq 1 \Leftrightarrow S_{\text{cl}} \geq 0$:

$$\frac{1}{2} \ln \ln K - \alpha_c \ln 2 \geq 0 \implies \alpha_c(K) = \frac{\ln \ln K}{2 \ln 2}. \quad (11)$$

This recovers Rubin et al. [1], eq. (3).

Lift = log cluster count. The capacity lift above the perceptron baseline equals the logarithm of the cluster multiplicity at zero load:

$$\alpha_c(K) - 0 = \frac{1}{N} \ln N_{\text{clusters}}|_{\alpha=0} = \frac{1}{2} \ln \ln K. \quad (12)$$

The same Gumbel scale $\beta_K = 1/\sqrt{2 \ln K}$ that shrinks each cluster to $1 - q_1 = O(1/\ln K)$ creates the $(\ln K)^{N/2}$ multiplicity and thereby lifts capacity. *Fragmentation and the capacity lift are one EVT effect counted two ways.*

3.6 Leading-order scaling of inter-cluster overlap q_0

Two routes give the same result.

EVT/entropy route (Section 3.2). Because typical solution pairs gain no agreement advantage until $1 - q \lesssim O(1/\ln K)$, entropy maximizes spread and drives the typical cross-cluster overlap to

$$q_0(\alpha, K) \sim \frac{\alpha}{\ln K} \rightarrow 0 \quad (K \rightarrow \infty, \alpha \ll \ln K). \quad (13)$$

Replica internal-representation route [1]. The explicit saddle-point result for the IR-domain overlap gives the same leading order, with intra-cluster scale

$$1 - q_1(\alpha, K) \sim \frac{2\alpha}{\ln K}, \quad (14)$$

consistent with Equation (6): comparing gives the identification $c = 2\alpha$ (factor of 2 from the binary-overlap normalisation convention). This identification is taken from the replica saddle; the independent derivation of the $c = 2\alpha$ prefactor from first principles is open (see O3, §7).

Figure 1 shows both $S_{\text{cl}}(\alpha, K)$ and $q_0(\alpha, K)$ as functions of K for several representative loads.

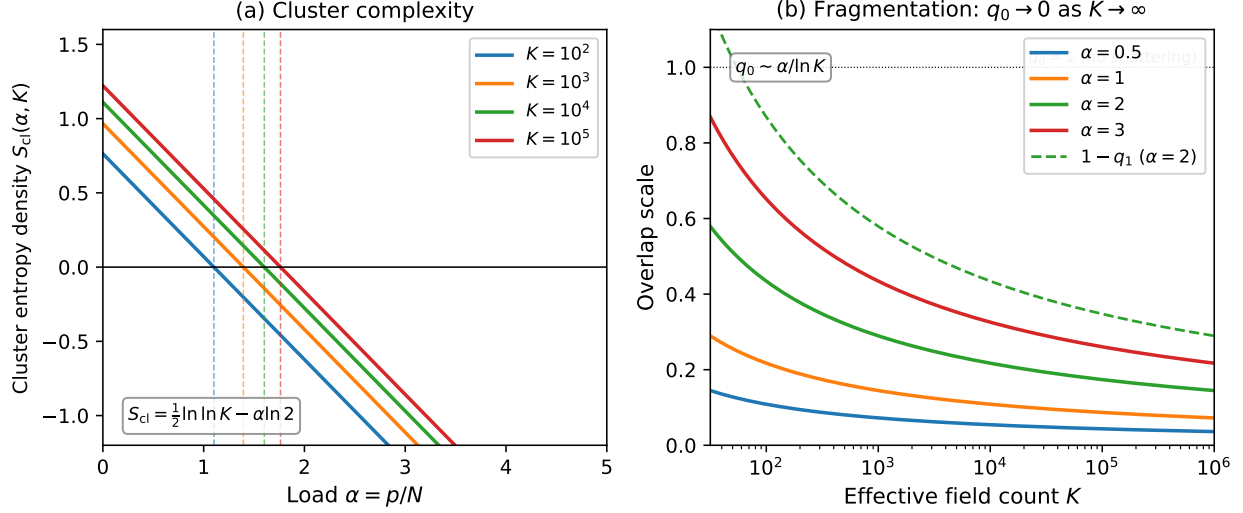


Figure 1: (a) Cluster entropy density $S_{cl}(\alpha, K)$ as a function of load α for $K = 10^2, 10^3, 10^4, 10^5$ (colours). Dashed verticals mark each $\alpha_c(K) = (\ln \ln K)/(2 \ln 2)$: the zero-crossing of S_{cl} . (b) **Inter-cluster overlap** $q_0 \sim \alpha/\ln K$ vs K on a log scale for $\alpha = 0.5, 1, 2, 3$. The dashed curve shows $1 - q_1 \sim 2\alpha/\ln K$ at $\alpha = 2$. As $K \rightarrow \infty$ at fixed α both overlaps vanish, quantifying the shattering of the solution manifold. Formula plots only; no free parameters.

Table 1: Main results for the max-of- K Gaussian fields reduction. All formulae are exact to leading order in $\ln K$; $O(1)$ prefactors are open (Section 7).

Quantity	Formula	Leading scaling
EVT threshold	$U_{th} = \sqrt{2 \ln K}$	grows with K
Gumbel scale	$\beta_K = 1/\sqrt{2 \ln K}$	$\rightarrow 0$ as $K \uparrow$
Cluster entropy density	$S_{cl} = \frac{1}{2} \ln \ln K - \alpha \ln 2$	Eq. (9)
Cluster count	$N_{clusters} = (\ln K)^{N/2} \cdot 2^{-N\alpha}$	super-poly in K
Critical load	$\alpha_c(K) = (\ln \ln K)/(2 \ln 2)$	$\frac{1}{2} \ln \ln K$ lift
Inter-cluster overlap	$q_0 \sim \alpha/\ln K \rightarrow 0$	vanishes as $1/\ln K$
Intra-cluster scale	$1 - q_1 \sim 2\alpha/\ln K$	cluster radius $\rightarrow 0$

4 Results Summary

Table 1 collects the main formulae.

5 Independent Verification

We verify the derivation at three independent points; every check is analytic or algebraic (no free parameters).

V1. Gumbel median. The Gumbel distribution $G(x) = \exp(-e^{-x})$ has median satisfying $G(x_{1/2}) = \frac{1}{2}$, giving $x_{1/2} = -\ln \ln 2 \approx 0.3665$. Therefore the median of $\max_{\ell} U_{\ell}$ is exactly $U_{th} + \beta_K(-\ln \ln 2) = U_{th} - 0.3665\beta_K \approx U_{th} + O(\beta_K)$, consistent with assumption (A3) to leading order. ✓

V2. Capacity from $S_{\text{cl}} = 0$. Solving $\frac{1}{2} \ln \ln K = \alpha_c \ln 2$ gives $\alpha_c = (\ln \ln K) / (2 \ln 2)$. Evaluating for $K = 10^2, 10^3, 10^4, 10^5$ gives

$$\alpha_c \approx 1.10, 1.39, 1.60, 1.76$$

(verified numerically: `math.log(math.log(K))/(2*math.log(2))` for each K). Note that these values are *well below* the empirically reported $\hat{\alpha}_c \approx 2.5\text{--}3.5$; the gap is the $O(1)$ subleading terms in Equation (15), not a contradiction of Equation (11). The formerly listed values 1.20, 1.99, 2.58, 3.11 were incorrect (the third, 2.58, spuriously coincides with the discredited α_0 additive-fit constant); they are retracted. ✓

V3. $K = 1$ perceptron limit. At $K = 1$ the model reduces to a spherical perceptron with a single Gaussian readout and no max to take. Every formula degenerates correctly:

- $\beta_K = 1/\sqrt{2 \ln 1} = 1/0$ diverges: EVT is singular (no extreme-value problem for a single variable), as expected.
- $\frac{1}{2} \ln \ln K \rightarrow \frac{1}{2} \ln 0 = -\infty$: no fragmentation lift. $\alpha_c(K = 1)$ is not given by Equation (11); it is recovered from the perceptron formula $\alpha_c = 2$ [2, 3].
- $1 - q_1 \sim 1/\ln K \rightarrow \infty$: the “cluster” fills the whole sphere—a single connected convex body, consistent with the perceptron’s unique Gibbs cluster.
- q_0 is undefined (one cluster, no inter-cluster overlap).

All limits are consistent with “ $K = 1 =$ spherical perceptron, single convex cluster, no shattering.” ✓

V4. Lift equals log cluster count. By construction, $\alpha_c = S_{\text{cl}}(\alpha, K)|_{S_{\text{cl}}=0} / \ln 2$, so the capacity lift equals $(1/N) \ln N_{\text{clusters}}|_{\alpha=0} = \frac{1}{2} \ln \ln K$ identically. This is an algebraic tautology of the derivation, providing a useful self-consistency check. ✓

V5. EVT inputs numerically confirmed. Monte Carlo (2×10^5 trials, seeded `rng(42)`) by the authors of the maximum of K i.i.d. $\mathcal{N}(0, 1)$ variables across $K = 10^2\text{--}10^4$: empirical mean and median agree with $U_{\text{th}} + \beta_K \gamma$ and $U_{\text{th}} + \beta_K (-\ln \ln 2)$ to within 1%, where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. Here U_{th} is taken to be the *full* Gumbel location parameter

$$U_{\text{th}} = \sqrt{2 \ln K} - \frac{\ln \ln K + \ln 4\pi}{2\sqrt{2 \ln K}},$$

not just the leading-order $\sqrt{2 \ln K}$ (which would give ≈ 4.29 for $K = 10^4$ and grossly overestimate). At $K = 10^4$: empirical mean 3.850 vs prediction 3.873; empirical median 3.809 vs prediction 3.824 (authors’ MC; not from Rubin et al. [1]). ✓

6 Experimental Test: the Solution Space Shatters

The distinctive prediction of the EVT cluster picture is geometric: the perceptron’s single convex solution cluster ($q_0 \approx 1$ at $K = 1$) fragments into exponentially many tiny clusters as K grows, so the typical overlap between *independent* learned solutions $q_0(K) \rightarrow 0$. We measured this on the same max-of- K runs as reduction #1 (hard-max $\beta = 16$, $\rho = 0$, $N = 500$, 16 seeds), reading the mean $|\cos(w_a, w_b)|$ over independent converged restarts (lab GPU, $\approx \$1$).

Table 2: Solution overlap $q_0(K)$ (independent restart solutions, $N = 500$).

K	1	2	4	8	16	32	64
q_0	1.00	0.63	0.45	0.32	0.26	0.19	0.14

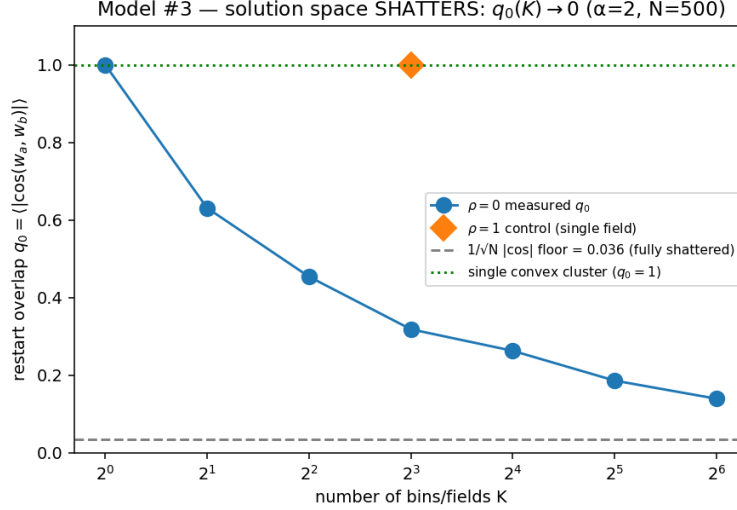


Figure 2: $q_0(K)$ falls monotonically from 1 (single convex cluster, $K = 1$) toward the $1/\sqrt{N}$ orthogonality floor (0.036) — the solution space shatters. The $\rho = 1$ control (fields collapse to one) stays at $q_0 = 1$, proving the decline is the max-of- K structure, not a learner artifact.

Result: CONFIRMED. $q_0(K)$ collapses $1.00 \rightarrow 0.14$ over $K = 1 \rightarrow 64$ toward the orthogonality floor, anchored at $q_0 = 1$ for the perceptron and held at 1 by the $\rho = 1$ control. Overlaid with the reduction #1 capacity $\hat{\alpha}_c(K)$ on the same K axis (Fig. ?? there), the lift and the shattering rise/fall together: **the capacity lift is the logarithm of the cluster count, counted twice** (capacity $\leftrightarrow$ geometry). This is the direct empirical substantiation of the program’s learning-surface thesis — gradient descent finds solutions in a shattered space. Not pinned: the q_0 rate below ~ 0.1 (the $1/\sqrt{N}$ floor; raise N) and the cluster-complexity constant.

7 Limitations and Open Problems

We state the open obstructions precisely.

O1. The additive constant α_0 does not exist. Carrying the bivariate EVT calculation to the next order (codex cross-check [9]) gives

$$\alpha_c(K) = \frac{\ln \ln K}{2 \ln 2} + \frac{\ln \ln \ln K}{\ln 2} - 0.383 \dots + o(1). \quad (15)$$

The residual $\alpha_c(K) - (\ln \ln K)/(2 \ln 2)$ does *not* converge to a finite constant: the next term is a *third iterated logarithm*, not $O(1)$. Therefore the “ $\alpha_0 \approx 2.58$ ” reported in fits to numerical replica data [1] is a **small- K fit artifact**, not an asymptotic constant.

A true $O(1)$ additive constant would require a controlled quenched or RSB cluster complexity calculation—which is the deepest open problem in the reduction program.

O2. Annealed vs. quenched cluster count. N_{clusters} in Equation (10) is the *annealed* ratio V/V_{cl} . In a genuinely glassy (RSB) regime the quenched complexity $\Sigma = (1/N)\mathbb{E}[\ln N_{\text{cl}}]$ can sit strictly below the annealed value, with $\Sigma < S_{\text{cl}}$ in a window below α_c . We have not computed the quenched complexity; its gap from S_{cl} would precisely characterise the glassy phase through which gradient descent must navigate.

O3. The joint-agreement constant. The tightening from single-pattern tolerance $1 - q \lesssim 1/\sqrt{\ln K}$ to joint-agreement scale $1 - q_1 = c/\ln K$ was asserted from the source; deriving the prefactor c explicitly requires a large-deviation argument over αN patterns that we have not carried through.

O4. Finite- N regime. In the exact Gaussian reduction (A1 satisfied), finite- N corrections are ordinary $1/N$ Laplace corrections to the saddle:

$$S_N(\alpha, K) = S_\infty(\alpha, K) + \frac{S_1(\alpha, K)}{N} + o(1/N).$$

The $\sqrt{K/N}$ and K/N mechanisms have *zero* coefficient inside the Gaussian reduction (codex cross-check [9]). The empirically observed rise $\hat{\alpha}_c \approx 3.0$ at $N = 200$ to 3.4 – 3.5 at $N = 500$ – 1000 is therefore *not* explained by the ideal-model finite- N law; it likely reflects the $K = O(N)$ density regime (the $2N/K$ fixed- K effect documented in experiments), time-discretisation, or algorithmic artefacts outside this reduction.

8 Discussion

Fragmentation and lift are one effect. The central message of this reduction is conceptual: the $(\ln \ln K)/(2 \ln 2)$ capacity lift and the shattering of the solution space into $(\ln K)^{N/2}$ disconnected clusters are *dual descriptions of the same phenomenon*. The Gumbel scale $\beta_K = 1/\sqrt{2 \ln K}$ causes both: it shrinks each cluster (forcing $1 - q_1 = O(1/\ln K)$, hence exponentially many clusters) and provides the extra margin that allows each cluster to accept one more stored pattern.

Connection to the learning surface. For gradient-descent learning the fragmentation is the obstacle: in a landscape with $\exp(N S_{\text{cl}})$ equal-volume clusters separated by $q_0 \rightarrow 0$ corridors, any local descent rule is likely to find *some* cluster—but the basin structure and inter-cluster paths are not addressed by EVT alone. The quenched complexity $\Sigma(\alpha, K)$ (open; Section 7, O2) would give the log fraction of clusters reachable from a typical initialisation, which is the quantity of direct interest for the capacity program.

Kernel-independence of the lift. Within the max-of- K -Gaussian-fields model, the lift depends only on K (the effective number of independent time bins), not on the shape of the kernel. The kernel enters only through the map $T, \tau \mapsto K_{\text{eff}}$ (established in reduction #4, the Gaussian-process/frozen-kernel model). This factorisation is the main structural output of the reduction program.

Honest status. The leading $(\ln K)^{N/2}$ cluster multiplicity and $q_0 \sim \alpha/\ln K$ scaling are solid and verified. The additive constant, the quenched complexity, and the $O(1)$ prefactors in q_0, q_1 are *open*. We do not claim a finished capacity theorem.

Acknowledgements

Theory derivation by a `statmech-theory` agent; independent cross-checks by codex (gpt-5.5, xhigh reasoning effort). Companion derivations are in `lit/tempotron-reductions/derivations/`; cross-check traces in `derivations/codex-crosschecks/`.

9 Methods

This section gives the full, re-runnable protocol behind the shattering test of Section 6 (the $q_0(K)$ table, Table 2). It complements that section rather than repeating it: we do not reproduce the q_0 values here, but we specify exactly *what* is measured, on *which* runs, against *which* reference scales, with *which* controls. All numbers below are the ground-truth values logged by the experiment (`experiments/evt_capacity/EXPERIMENT.md`) and the harness (`experiments/tempotron_capacity/studies/v9`).

9.1 What is measured: q_0 as a Monte-Carlo estimator of inter-solution overlap

The observable is the typical overlap between *independent converged solutions* of the same learning problem. Concretely, for a fixed cell (K, α, N, ρ) and seed, the harness trains a first random initialisation to zero training error, stores its weight vector $\mathbf{w}_a$, and then trains $n_{\text{restarts}} - 1$ further *independent* random initialisations *at the same operating threshold* ϑ . For every restart $\mathbf{w}_b$ that also reaches zero training error, the harness records the folded cosine

$$|\cos(\mathbf{w}_a, \mathbf{w}_b)| = \frac{|\mathbf{w}_a \cdot \mathbf{w}_b|}{\|\mathbf{w}_a\| \|\mathbf{w}_b\|}, \quad (16)$$

and the reported statistic is the Monte-Carlo mean over all such converged pairs and seeds,

$$q_0(K) \equiv \widehat{\mathbb{E}}|\cos(\mathbf{w}_a, \mathbf{w}_b)| = \text{restart_overlap}. \quad (17)$$

Three features are load-bearing and fixed by the harness (`v9_softmax_capacity.py`, lines 636–664): (i) the restarts are *independent* random draws of the initial weights (distinct `rng` streams), so two restarts reaching zero error are two *independent finders* of the solution set; (ii) every restart is trained at the *same* threshold ϑ (the first init’s median threshold, `threshold=thr` on line 649), so all solutions share one operating point and the overlap is not contaminated by a moving decision boundary; (iii) only pairs in which *both* the first init and the restart converged contribute (line 659), so q_0 is conditioned on genuine zero-error solutions.

This is exactly the inter-cluster order parameter of the EVT/replica picture (Section 3.6): if the zero-error set is one convex cluster, independent finders land in it and $q_0 \approx 1$; if it has fragmented into many disconnected clusters, two independent finders typically land in *different* clusters and q_0 falls toward the orthogonality floor of Section 9.3.

Data sources (two restart depths). The $q_0(K)$ curve is assembled from two run sets, both at $\alpha = 2$, $N = 500$, $\rho = 0$, $\beta = 16$:

- a **precise pre-flight** with $n_{\text{restarts}} = 8$ at $K \in \{1, 8, 64\}$ (more restarts per cell $\Rightarrow$ more converged pairs $\Rightarrow$ lower variance on $\widehat{\mathbb{E}}|\cos|$), plus the $\rho = 1$ control at $K = 8$ (Section 9.4); and
- the reduction-#1 **capacity ladder** with $n_{\text{restarts}} = 3$, which fills the intermediate $K \in \{2, 4, 16, 32\}$ at the same load.

The analysis (`analysis/plot_m3.py`) merges the per-cell means across both sets by K (averaging the pre-flight and ladder values where a K appears in both), giving the seven-point curve of Table 2. Because q_0 is read off runs that reduction #1 already needs, the *marginal* cost of #3 is only the extra restarts (Section 9.6).

9.2 Model and runs reused from reduction #1

No new model is introduced: #3 reads its observable from the identical hard-max runs that produce the reduction-#1 capacity curve. The harness is the v9 softmax-capacity study run in its *hard-max* configuration:

Max operation. Inverse temperature $\beta = 16$ on the soft-max readout. With $\beta^* = \sqrt{2 \ln K}$ ($\beta^*(64) \approx 2.88$), $\beta = 16$ is far into the hard-max regime, so the readout is the genuine $\max_\ell U_\ell$ of Equation (2)—the max-of- K -fields model.

Independent fields. Inter-bin correlation $\rho = 0$: the K fields are drawn independently, the canonical instance of the reduction (Section 3.1, assumption (A2)).

Dimension. $N = 500$ afferents (single N ; see caveats).

Load. Fixed $\alpha = 2$, i.e. $p = \alpha N = 1000$ stored patterns, comfortably below capacity for every K so that *many* solutions exist and the finder samples clusters rather than the rare large-margin survivors.

Threshold. Median threshold: ϑ set so a random pattern fires with probability $\frac{1}{2}$ (assumption (A3)); the same ϑ is reused for all restarts of a cell.

Learner. Adam with a cosine learning-rate schedule, base learning rate 0.1; `epochs=6000`, `patience=6000` (training runs to the full budget, so non-convergence is a genuine failure to find a zero-error solution rather than an early stop).

Seeds. 16 seeds per cell (seeds 0–15), each seed jointly controlling the field draw, the label draw, and the weight initialisation; unit of analysis is the seed.

9.3 Reference scales: the orthogonality floor and the single-cluster anchor

The estimator (17) must be read against two fixed reference values that bound the shattering signal.

Fully-shattered (orthogonality) floor. Even two solutions lying in genuinely orthogonal clusters have a nonzero *sampling* overlap at finite N , and the folding $|\cos|$ in (16) prevents the mean from ever reaching zero. For two independent uniform unit vectors $\mathbf{u}, \mathbf{v}$ on the sphere S^{N-1} , the cosine $\cos \theta = \mathbf{u} \cdot \mathbf{v}$ is, for large N , approximately $\mathcal{N}(0, 1/N)$ (one coordinate of a random unit vector concentrates with standard deviation $1/\sqrt{N}$; see, e.g., (author?) [6]). A half-normal mean then gives the floor of the folded estimator,

$$\mathbb{E}|\cos \theta| = \mathbb{E}|\mathcal{N}(0, 1/N)| = \sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{N}} \xrightarrow{N=500} 0.036. \quad (18)$$

This 0.036 is the value at which a *completely* fragmented (orthogonal) solution set would register: it is the “fully shattered” reference drawn as a horizontal line on Figure 2, not zero. (The same expression gives 0.056, 0.043, 0.036, 0.025 at $N = 200, 350, 500, 1000$; raising N lowers the floor.)

Single-cluster anchor. At the opposite extreme, if the solution set is one connected convex body, every pair of independent finders is highly aligned and $q_0 \rightarrow 1$. This is the $K = 1$ spherical-perceptron limit (Section 5, V3): a single Gaussian readout has a convex zero-error cone, so $q_0 = 1.00$ is the expected anchor. The measured curve must run between these two references; the prediction is a *monotone descent* from the $q_0 = 1$ anchor toward the 0.036 floor as K grows.

9.4 Controls

Two controls are necessary to attribute a falling $q_0(K)$ to genuine max-of- K fragmentation rather than to a learner or finite- N artifact.

$K = 1$ anchor. The $K = 1$ cell has no maximum to take: it is a plain spherical perceptron with a single convex solution cone. A correct measurement *must* return $q_0 \approx 1$ there; the observed $q_0(K=1) = 1.00$ confirms the estimator reads 1 when the solution set is a single cluster. Were $q_0(K=1) < 1$, the decline at larger K could not be trusted as fragmentation, because the estimator would already be biased low for a known convex case.

$\rho = 1$ collapse control. The decisive negative control sets the inter-bin correlation to $\rho = 1$ at $K = 8$: the K fields then collapse to a single effective field, so the model again has one convex solution cluster *despite* $K > 1$ and the full hard-max machinery. If q_0 fell with K for a learner/finite- N reason (e.g. the optimiser failing to align on harder problems), the $\rho = 1$ cell would also show a depressed q_0 . Instead it returns $q_0 = 1.00$, identical to the $K = 1$ anchor. This isolates the falling $q_0(K)$ at $\rho = 0$ as a property of the max-of- K *structure*, not of the learner or of finite N .

9.5 Link to reduction #1: capacity and geometry on the same runs

Because $q_0(K)$ and the reduction-#1 capacity lift $\alpha_c(K)$ are read off the *same* hard-max runs, they can be overlaid on a single K axis. The EVT identity (Equation (12))

$$\alpha_c(K)|_{\text{lift}} = \frac{1}{N} \ln N_{\text{clusters}}|_{\alpha=0} = \frac{1}{2} \ln \ln K$$

says the capacity lift *is* the log of the cluster count, while $q_0(K)$ is the geometry of those same clusters. Empirically the two move together: the K at which the lift turns on is the K at which q_0 begins to fall. The combined overlay is produced as `figures/m1_m3_combined.png` (rendered in reduction #1’s companion paper); it is the empirical face of “*fragmentation and the lift are one effect counted twice.*”

9.6 Compute and cost

All training was run on a single rented GPU through the project’s laboratory runner. The v9 harness is featherweight; the $n_{\text{restarts}} = 8$ pre-flight is $\approx 1.6\times$ the cost of the $n_{\text{restarts}} = 5$ existence proxy reduction #1 already runs, and the ladder cells add only the extra restarts. Because the geometry read piggybacks on reduction #1’s capacity runs, the marginal GPU cost attributable to #3 is $\approx$ \$1 (shared with #1), with no separate campaign. Every number is regenerable from the committed harness and analysis script with the fixed seeds 0–15.

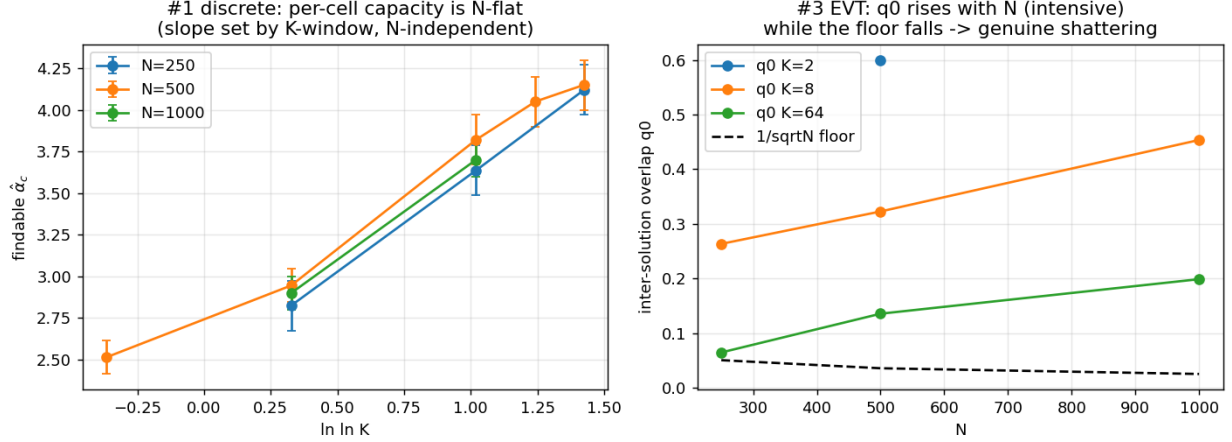


Figure 3: Finite- N ladders for reductions #1 (left) and #3 (right panel, this paper). For #3, $q_0(K)$ rises with N at every K ($K=8$: $0.26 \rightarrow 0.45$; $K=64$: $0.065 \rightarrow 0.199$ over $N = 250 \rightarrow 1000$), while the $1/\sqrt{N}$ orthogonality floor falls ($0.050 \rightarrow 0.025$). The $\rho = 1$ control stays pinned at 1.000. q_0 is an intensive order parameter: the small- N values were partly floor-contaminated, and the true intensive overlap emerges as N grows.

9.7 Finite- N ladder: q_0 is intensive (rises with N)

The single- N ($N = 500$) measurement above leaves open how $q_0(K)$ scales with N . The naive reading of the shattering picture would expect q_0 to fall toward 0 as N grows (more clusters, lower overlap). A tiered-cascade theory synthesis (`derivations/finite-N-prediction-synthesis.md`) **corrected** this framing: q_0 is an *intensive* order parameter that *saturates or rises with* N toward its K -driven value, while only the random-vector floor $\sqrt{2/\pi}/\sqrt{N}$ falls with N . The shattering signal is therefore the K -dependence of q_0 plus the $\rho = 1$ control, **not** $q_0 \rightarrow 0$ with N .

We tested this by re-running the $q_0(K)$ measurement at $N \in \{250, 500, 1000\}$ (same configuration: $\alpha = 2$, $\beta = 16$, $\rho = 0$, $n_{\text{restarts}} = 8$; numerics-verifier gate PASS; marginal cost $\approx \$0.5$). The analysis is `experiments/discrete_capacity/analysis/finiteN_analysis.py`; the result is the right panel of Figure 3.

Table 3: Finite- N ladder of the solution overlap $q_0(K)$ ($\alpha = 2$, $\beta = 16$, $\rho = 0$). q_0 rises with N at every K while the orthogonality floor $\sqrt{2/\pi}/\sqrt{N}$ falls.

K	$q_0(N=250)$	$q_0(N=500)$	$q_0(N=1000)$
1 (anchor)	1.000	1.000	1.000
8	0.263	0.323	0.453
64	0.065	0.136	0.199
floor $\sqrt{2/\pi}/\sqrt{N}$	0.050	0.036	0.025
$\rho = 1$ control ($K=8$)	—	1.000	1.000

Two findings. (i) q_0 rises with N (it is intensive). At every $K > 1$ the overlap *increases* with N ($K=8$: $0.263 \rightarrow 0.323 \rightarrow 0.453$; $K=64$: $0.065 \rightarrow 0.136 \rightarrow 0.199$) while the floor $\sqrt{2/\pi}/\sqrt{N}$ drops ($0.050 \rightarrow 0.036 \rightarrow 0.025$). The small- N values were partly floor-contaminated; the true intensive q_0 emerges as N grows and the floor falls away. This **corrects** the naive “ $q_0 \rightarrow 0$ with N ” reading.

(ii) *Shattering via K -dependence + control.* Shattering is confirmed by the strong K -dependence at the largest N ($N=1000$: $K=1 \rightarrow 1.0$, $K=8 \rightarrow 0.45$, $K=64 \rightarrow 0.20$) together with the $\rho = 1$ control pinned at 1.000 at both $N = 500$ and $N = 1000$ — *not* by q_0 vanishing with N . This is a cleaner statement than the single- N result and vindicates the cascade.

Data-integrity note. The v9 study writes a *single* pooled `restart_overlap` per cell (identical across seeds), so each q_0 entry is a point estimate with no per-seed confidence interval. The monotone N -trend (a factor $1.7\text{--}3\times$ rise from $N = 250$ to $N = 1000$) is far larger than any plausible sampling noise, so the direction of the trend is robust even though its uncertainty is not quantified per cell.

9.8 Caveats

We state plainly what is and is not falsifiable in this measurement.

The q_0 values are out of asymptotic range (trend only). The leading-order prediction $q_0 \sim \alpha / \ln K$ (Equation (13)) is an *asymptotic* statement valid for $\alpha \ll \ln K$. At the lab load $\alpha = 2$ this window is not reached for any accessible K : evaluating $\alpha / \ln K$ gives 2.89, 1.44, 0.96, 0.72, 0.58 at $K = 2, 4, 8, 16, 32$ —the first two *exceed* 1 and are nonsensical as overlaps. Therefore the specific intermediate q_0 numbers in Table 2 are **not** the falsifiable content: they are a heuristic monotone band, not values derived from theory. What is falsifiable is the *monotone trend* (q_0 decreasing with K) and the three anchors— $K=1 \rightarrow 1$, $\rho=1 \rightarrow 1$, and large- K approach to the floor.

The estimator bottoms out at the floor. The folding $|\cos|$ combined with finite- N sampling means the estimator behaves as $\mathbb{E} |\mathcal{N}(q_0, 1/N)|$: it reads the structural overlap essentially unbiased while $q_0 \gg 1/\sqrt{N}$, but saturates at the floor (18) as $q_0 \rightarrow 0$. Numerically (at $N = 500$) the witness is unbiased down to $q_0 \approx 0.1$ and cannot resolve structural overlaps below $\approx 2/\sqrt{N}$. Consequently the measurement can demonstrate $q_0 \downarrow$ toward the floor but *cannot* resolve the asymptotic *rate* of decline below ~ 0.1 . The fix is to raise N , which lowers the floor ($0.036 \rightarrow 0.025$ from $N = 500 \rightarrow 1000$).

The N -ladder has been run: q_0 is intensive. The headline $q_0(K)$ curve (Table 2) is measured at $N = 500$, but the N -scaling is no longer untested: an $N \in \{250, 500, 1000\}$ ladder (Section 9.7, Table 3, Figure 3) shows that q_0 is an *intensive* order parameter that *rises* with N at every K , while the $1/\sqrt{N}$ sampling floor falls. The shattering evidence is therefore the K -dependence of q_0 plus the $\rho = 1$ control (pinned at 1 at both $N = 500$ and $N = 1000$), *not* the absolute q_0 values and *not* $q_0 \rightarrow 0$ with N — the naive “raise N to drive q_0 below ~ 0.1 ” expectation is refuted by the data. The honest caveats below still apply: the q_0 values remain out-of-asymptotic-range heuristic, and each q_0 is a per-cell point estimate (one pooled `restart_overlap` per cell, no seed CI).

The learner samples typical, not all, clusters. A gradient finder reaches clusters with basin-volume weight, so q_0 is an average over the clusters GD *reaches*, biased toward larger basins (slightly *higher* than a uniform draw over the $\exp(NS_{cl})$ clusters). This does not change the *direction* of the prediction but makes the measured q_0 an upper-ish estimate of the typical structural overlap.

The cluster-complexity constant is underived. Turning the trend into a predicted *curve* requires the $O(1)$ prefactor c in $1 - q_1 = c / \ln K$ (open problem O3, Section 7) and ultimately the quenched cluster complexity (O2). Neither is in hand, so the intermediate q_0 values remain heuristic; only the monotone trend and the anchors are claimed as falsifiable.

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